Name: Ashik Ahmed Sajib
ID: IT24648

Question: Is set of odd numbers with binary operations (+), ie $\langle O, + \rangle$ an abelian group If not explain the reasons with necessary notations.

Solution:

$O = \{ \ldots -3, -1, 1, 3, 5, 7, 9, 11, 13, 15 \ldots \}$

1. Clousure: if $a, b \in O$ then $a+b \in O$

Let $a = 3$
$b = 5$

then $3+5 = 8$ not $\in O$

So, condition fails this requirement

2. Identify element: 0 is even number, so

in $e+a \in G$ is work out,

3. Associative: If $\forall a, b, c \in O$ then

$a+(b+c) = (a+b)+c \in O$

Let $a = 3$
$b = 5$
$c = -5$

then $3+(5-3) = (3+5)-3 = 5 \in O$

So, this condition is satisfying.

4. <u>Inverse Element</u>: If $a, a' \in O$ and $(a+a') = e \in O$

Let $a = 3$

$a' = -3$

then, $3-3 = 0 \in O$

So, the set is not satisfying the inverse property.

5. <u>Communicative</u>: If $\forall a, b \in O$, the $(a+b) = b+a \in O$

Let $a = 5$, $b = 7$

then, $5+7 = 7+5 = 12$ not $\in O$

So, this condition is satisfying.

Here, given odd numbers set is not an abelian because it is not pass the all condition of abelian.

CamScanner